Department of Physics, FNSPE, CTU in Prague

# 02YMECH – Mechanics

## Midterm 1

6.11.2026 / 18:00–19:20

**Q1:** A small block of mass $m$ moves along a frictionless horizontal surface. The block is subjected to a position-dependent force acting along the $x$-axis,

$$F(x) = -\frac{A}{(x+a)^3},$$

where $A > 0$ and $a > 0$ are constants, and the motion is restricted to $x \geq 0$. Initially, the block is at $x(0) = 0$ and moves in the positive $x$-direction with velocity

$$v(0) = v_0 = \sqrt{\frac{A}{ma^2}}.$$

**(a)** Write down the equation of motion for the block. – *0.5 points*

**(b)** Determine the velocity of the block as a function of its position $x$. – *0.5 points*

Hint: $\ddot{x} = \dfrac{dv}{dx}\dfrac{dx}{dt}$

**Q2:** A particle of mass $m$ moves on a frictionless horizontal surface, along the inner side of a fixed circular wall of radius $R$. The particle is attached to a spring of negligible mass, spring constant $k$, and natural length $L < R$, whose other end is fixed at the center $O$ of the wall. The particle executes uniform circular motion with constant angular speed $\omega$, while remaining in contact with the wall.

**(a)** Determine the magnitude $N$ of the normal force exerted by the wall on the particle as a function of $m$, $k$, $L$, $R$, and $\omega$. – *0.8 points*

**(b)** Determine the minimum angular speed $\omega$ required for the particle to remain in contact with the wall. – *0.2 points*

**Q3:** The displacement of a damped oscillator is given by $x(t) = e^{-\beta t} x_0 \cos(\omega_1 t)$. How many oscillations $N$ does it approximately take for the maximum potential energy stored in the oscillator to reduce to $1/2$ of its initial value? Express your result in terms of the quality factor $Q$,$\ln 2$, and $\pi$. Assume that the oscillator is weakly damped, i.e., $\omega_1 \approx \omega_0$. (Givens: $Q = \omega_0/(2\beta)$, $T_0 = 2\pi/\omega_0$) — *1 point*

**Q4:** A block of mass $m$ slides down a rough inclined plane making an angle $\theta$ with the horizontal. The block travels a distance $d$ along the incline, during which its speed decreases from $v_1$ to $v_2 < v_1$. Assuming that the friction force is constant, determine the magnitude of this force. — *1 point*